

COURSE TITLE : MATERIALS FOR PRINTING
COURSE CODE : 3103
COURSE CATEGORY : B
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Printing inks	22
2	Paper substrates	23
3	Plastic substrates	20
4	Other specialized substrates and materials	10
TOTAL		75

COURSE OUTCOME :

MODULE	G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	1	To comprehend about different aspects of printing inks
2	2	To Comprehend the different aspects of paper and board
3	3	To understand different characteristics of plastics used for printing
4	4	To understand various characteristic of miscellaneous printing materials

MODULE I PRINTING INKS

1.1.0. To comprehend about different aspects of printing inks

- 1.1.1. To describe the Ingredients of ink
- 1.1.2. To explain the Properties of ink
- 1.1.3. To differentiate the Ink drying mechanisms
- 1.1.4. To know ink manufacturing process.
- 1.1.5. To identify the Characteristics of lithographic (offset) inks
- 1.1.6. To identify the Characteristics of gravure printing inks
- 1.1.7. To identify the Characteristics of flexographic inks
- 1.1.8. To identify the Characteristics of screen printing inks
- 1.1.9. To distinguish different types of inks
- 1.1.10. To state special color inks

MODULE II PAPER SUBSTRATES

2.1.0 To comprehend the different aspects of paper and board

- 2.1.1 To know paper and board raw materials.
- 2.1.2 To describe paper manufacturing process
- 2.1.3 To distinguish types of paper
- 2.1.4 To list out types of paper and their use
- 2.1.5 To list out different paper coatings

- 2.1.6 To distinguish between paper and board
- 2.1.7 To identify the types of board
- 2.1.8 To list out different standard size of paper and board
- 2.1.9 To explain the different properties of paper and Board
- 2.1.10 To explain different printability properties of paper

MODULE III PLASTIC SUBSTRATES

3.1.0 To understand different characteristics of plastics used for printing

- 3.1.1 To define plastic materials
- 3.1.2 To identify the different types of plastics
- 3.1.3 To describe the different properties of the plastics
- 3.1.4 To identify the printability properties of plastics
- 3.1.5 To define different packaging properties of plastics

MODULE IV OTHER SPECIALIZED SUBSTRATES AND MATERIALS

4.1.0 To understand various characteristic of miscellaneous printing materials

- 4.1.1 To compare different types of adhesives
- 4.1.2 To explain properties of dampening solution
- 4.1.3 To identify the ingredients of dampening solution and their properties
- 4.1.4 To define blanket and its types
- 4.1.5 To define leather material
- 4.1.6 To state the different properties of glass
- 4.1.7 To list out the printing methods used on the glass
- 4.1.8 To define the aluminium foil
- 4.1.9 To list out different properties of metal foils
- 4.1.10 To state different printing method used on the metal foils

CONTENT DETAILS

MODULE I

Ink- Introduction, Components of ink-Pigments/ Dyes, Resins, solvents , additives-Plasticiser, Wax, drier, chelating agent, antioxidant, surfactants, alkali, defoamer. Ink properties: Color Properties, Flow properties- gloss, length, viscosity, tack, Working properties-. Ink drying and curing-oxidation, evaporation, penetration, Precipitation, radiation curing. Ink manufacturing steps (brief)- Varnish manufacture, pigment dispersal. Ink for different printing process- Offset, gravure, flexo, screen . Types of printing inks: Standard printing ink- Web Offset Non-Heatset (coldset) Inks, Web Offset Heatset Inks, Quickset Inks, Sheet-fed Inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High-Fidelity Ink. Specialty Ink- Non-porous Ink, Metallic Ink, Fluorescent Ink, magnetic ink, Scratch and Sniff Ink, edible ink, Moisture Resistant Ink, Security Ink. Names of standard ink manufactures used in India and world.

MODULE II

Introduction to paper and board. Raw materials for paper and board (brief). Pulping-various pulping process. Paper making process. Paper finishing- super calendering, coating, Mill conditioning. Different types of paper and their uses. Board making process. Different types of Board and their uses. Paper and Board sizes

Paper and Board Properties: printability properties , Runnability- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Abrasion Resistance, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness.

MODULE III

Plastic- definition, introduction. Classification of plastic (briefly)-Natural, synthetic , semi-synthetic , thermosetting, Thermoplastic. Different types of plastics (introduction and Properties)- PE, LDPE, LLDPE, HDPE, PP, PET,BOPP, Polystyrene, polyester etc. .Surface treatment on plastics (Briefly)- Surface Cleaning , Mechanical Treatment (Surface Roughening), Corona Treatment , Flame Treatment , Plasma Treatment, Chemical Etching .Plastic corrugated board.

MODULE IV

Adhesive: introduction, types. Dampening solution- ingredients and their properties. Blanket- definition and types of blankets. Leather- introduction, types of leathers. Glass- introduction, properties and application. Aluminium foil-definition, properties and application. Photographic and light sensitive materials – types. Metals for printing – types, characteristics. Plate treating solutions. Cleaning solutions.

Reference :		
Author	Title	Publishers
A S Porter	Manual for lithographic press operation	
C S Misra	Offset technology	Anupam prakashan
R. H. Leach	The Printing Ink Mannual 5 th Edition	Chapman &Hall, London
Fonaid E Tood	Printing inks	PIRA international united kingdom
Bob thomson	Printing Material Science and Technology	PIRA
W.H. Bureau	What the printer should know about the paper	GATF
Michael Bernard, John Peacock.	Handbook of printing and production	Springer Science & Business Media
Heigh. M. Speir	Introduction in Printing Technology	
Adams J.M, Faux,D.D and Rieber L.J	Printing Technology	Delmar Publishers, NewYork
Helmut Kipphan	Hand book of print media	Springer Science & Business Media